

install and configure everything PyDOS on the board. The setup sequence will prompt you for a few responses including board type, Wi-Fi ID/password, etc. The DOS-shell prompt reappears once the PyDOS is configured in its full glory. The setup sequence on my Pico 2 W is shown in Figure 4.

```
Adafruit CircuitPython 10.2.1 on 2026-05-13; Raspberry Pi Pico with rp2040
>>> import PyDOS
Starting Py-DOS...
Wild card length set to: 15

(137184)/>dir/w

Directory of /
[.]          [..]          [fsevents]  [lib]          [sd]
.metadata  never      .Trash-1000 .Trashes      boot_out.txt  code.py
PyDOS.py   settings.toml

  7 File(s)
  5 Dir(s)
    51892 Bytes.
  972800 Bytes free.

(122880)/>mem

    134.0 Kb free conventional memory
    24.0 Kb used conventional memory

(136992)/>date
The current date is: Sat 01/01/2000

(135872)/>time
The current time is: 12:01:37 AM

(134864)/>ver
PyDOS [Version 1.58]

(134256)/>prompt
PROMPT=$C$R$F$P$G

(133584)/>path
PATH=;/PyBasic

(132992)/>type code.py
print("Hello World!")
```

Figure 3. DOS-shell running commands

```
copy /cpython/boot.ro to /boot.ro
copy /cpython/boot.rw to /boot.rw
copy /cpython/code.py to /code.py
copy /cpython/fs.py to /fs.py
copy /cpython/main.py to /main.py
copy /cpython/matrix.py to /matrix.py
copy /cpython/playimage.py to /playimage.py
8 files(s) copied.
copy /cpython/lib/adafruit_connection_manager.mpy to /lib/adafruit_connection_manager.mpy
copy /cpython/lib/adafruit_dotstar.mpy to /lib/adafruit_dotstar.mpy
copy /cpython/lib/circuitpython_i2c_lcd.py to /lib/circuitpython_i2c_lcd.py
copy /cpython/lib/lcd_api.py to /lib/lcd_api.py
4 files(s) copied.
copy /cpython/lib/optional/neopixel.mpy to /lib/neopixel.mpy
1 files(s) copied.
*** Warning *** No board configuration file found
*** Warning *** /cpython/boardconfigs/pydos_bcfg_raspberry_pi_pico2_w.py
.
(N)anoConnect, (E) ESP32[C3/D4/S2/S3...] or (E) Pico-W board.
(N) Adafruit Fruit Jam,
(O)ther RP2040/STM32/Atmel SAMD/nRF/ARM Cortex-M board: E
copy /cpython/ESP32/autoexec.bat to /autoexec.bat
copy /cpython/ESP32/code.py to /code.py
copy /cpython/ESP32/main.py to /main.py
3 files(s) copied.
copy /cpython/ESP32/lib/adafruit_ntp.mpy to /lib/adafruit_ntp.mpy
copy /cpython/ESP32/lib/adafruit_requests.mpy to /lib/adafruit_requests.mpy
2 files(s) copied.
Are you using a Keyboard FeatherWing (Y/N)? N
Do you have a BB 010/020 I2C keyboard connected (Y/N)? N
Do you have a TFT FeatherWing Touch connected (Y/N)? N
Enter Wifi credentials now (/settings.toml file can be edited later) (Y/N)? N
Restarting PyDOS....
```

Figure 4: Full PyDOS setup

We are now ready to explore the additional functionality through the external programs we copied. Execute the command *blink* to start with onboard LED blinking and terminate its execution by typing *q*. Type *temperature* to keep printing the onchip sensor value and type *q* to stop the command. To scan the board I2C bus and display any device addresses you found, use the *i2cscan* command. To launch Python programs that require more memory than is available while running PyDOS, use the *runvm* command. Typing *bounce* will take you to the retro TUI animation days. You can configure a Wi-Fi ID/password (if skipped during the setup) with the *setup* command.

The full DOS-shell also provides various programming tools. You can launch a line editor or a full screen editor using the *edlin* and *edit* commands, respectively. A Python REPL in the shell can be launched using the *virtrepl* command. You can enjoy programming in good old Basic by executing *pybasic* commands. The full DOS-shell provides many other utilities including playing sound through the Pico GPIO, mounting an SD card to the file system, etc. Readers are encouraged to go through the PyDOS documentation to experience the full power of the DOS-shell.

To conclude, the pshell is good for quickly programming Pico boards in C. The PicoShell is a MicroPython layer that provides rich shell functionality with many useful commands. The PyDOS goes one step further to create popular classic DOS-like full environments on Pico boards. All of them are very useful for enhancing the end user experience with Pico boards. 

References

- Pshell project page, <https://github.com/lurk101/pshell>
- Pshell C examples, <https://github.com/lurk101/pshell/tree/master/c-examples>
- PicoShell project page, <https://github.com/patrickp02/PicoShell>
- PyDOS project page, <https://github.com/RetiredWizard/PyDOS>

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